

Week 4: Markov Chain Monte Carlo

Jaisel Singh (js6897)

October 7, 2026

What stood out to me in Neal's review was that Bayesian inference often comes down to averaging over plausible parameter values. This changes the goal from last week's optimization: a single high-probability point does not describe the uncertainty around it. MCMC constructs a Markov chain whose stationary distribution is the target posterior, allowing us to estimate expectations from its draws when the chain explores adequately. I found it especially useful that Metropolis acceptance ratios cancel the unknown normalizing constant. We can therefore work with an unnormalized posterior even when computing its integral is impractical.

The gap between a correct algorithm and a useful finite run also stood out. Successive draws are dependent, and strong positive autocorrelation means that collecting many draws may add little information. I found effective sample size a more meaningful way to judge precision than the raw iteration count. A small Monte Carlo standard error does not mean the posterior itself is narrow: one measures uncertainty in our numerical estimate, while the other reflects uncertainty about the model's parameters. This distinction makes me more careful about what it means to say that a result is precise.

Hoffman and Gelman's No-U-Turn Sampler addresses some of this computational difficulty. Hamiltonian Monte Carlo uses momentum and gradients to propose moves that can travel farther through the distribution than a random walk. NUTS chooses trajectory lengths automatically by stopping expansion when a path starts doubling back, while preserving the target distribution through its candidate-selection rules. I like how this makes a sophisticated sampler easier to use, though automatic tuning does not establish convergence. My remaining question is how to distinguish a chain that has explored the posterior well from one that looks stable because it is stuck near a single mode.

Readings: Neal (1993), *Probabilistic Inference Using Markov Chain Monte Carlo Methods*; Hoffman and Gelman (2014), *The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo*.